

Antiderivatives and Indefinite Integrals



Basic antiderivatives

1 Find each indefinite integral:

a $\int (3x^2 - 4x + 5) dx$

b $\int \frac{1}{x^2} dx$

c $\int \sqrt{x} dx$

d $\int \sec^2 x dx$

2 Find $\int (4\sin x + e^x) dx$.

Using initial conditions

3 Find $f(x)$ given that $f'(x) = 6x^2 - 2$ and $f(1) = 5$.

4 A particle moves along a line with velocity $v(t) = 3t^2 - 6t$ m/s. If its position at $t = 0$ is $s(0) = 4$ m, find its position function $s(t)$.

5 An object is dropped from rest, so its acceleration is $a(t) = -9.8$ m/s².

a Find the velocity function $v(t)$.

b If the object starts at height $s(0) = 45$ m, find its position function $s(t)$.

c Find how long it takes to hit the ground ($s = 0$), to two decimal places.

More antiderivatives

6 Find each indefinite integral:

a $\int (\frac{1}{x} - 3x^{-4}) dx$

b $\int (2\sec x \tan x) dx$

c $\int \frac{4}{\sqrt{1-x^2}} dx$

7 Find the general antiderivative of $f(x) = (x + 1)^3$ (expand first, or use the power rule on the whole expression, and state which method you used).

Motion applications

- 8 A rocket's acceleration is $a(t) = 6t \text{ m/s}^2$, with $v(0) = 0$ and $s(0) = 0$.
- a Find $v(t)$.
 - b Find $s(t)$.
 - c Find the distance traveled in the first 4 seconds.